Seyed Milad Hatamlee, Ph.D.

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EDUCATION

Ph.D., Mechanical Engineering *Jan. 2022 – May 2026*

University at Buffalo, State University of New York, Buffalo, NY

Dissertation: Thermodynamics and Transport of Small Molecules under Nanoconfinement.

Advisor: Dr. Jiaoyan Li.

M.S., Mechanical Engineering *Sep. 2017 – Feb. 2020*

Imam Khomeini International University, Qazvin, Iran

Thesis: Heat Transport in 2D Structures of Carbon Nitride by Non-Equilibrium Molecular Dynamics.

Advisor: Dr. Ali Rajabpour.

B.S., Mechanical Engineering *Sep. 2013 – Sep. 2017*

K.N. Toosi University of Technology, Tehran, Iran

Thesis: Comparison of Ventilation Systems for Large Commercial Projects from an Economic Perspective.

Advisor: Dr. Ali Behbahani Nia.

RESEARCH INTERESTS

Interfacial heat transfer and nanoscale thermal transport; confined-fluid flow and phase behavior; molecular thermodynamics; validated atomistic models for thermal-fluid systems.

TEACHING EXPERIENCE

Instructor of Record — MAE 204, Thermodynamics I *Summer 2024*

Department of Mechanical and Aerospace Engineering, University at Buffalo

- Held full instructional responsibility for a required undergraduate course serving 40+ students: syllabus, lecture delivery, assessment design, recitation sessions, grading policy, and final grade assignment.
- Received positive course evaluations for clarity of instruction on the first and second laws, thermodynamic cycles, equations of state, and energy conversion systems.
- Restructured recitation around worked-example scaffolding and low-stakes formative quizzes after identifying property-table interpolation and control-volume selection as the two dominant failure points.

Teaching Assistant — MAE 385, Engineering Materials Laboratory *Spring 2023 to 2026*

Department of Mechanical and Aerospace Engineering, University at Buffalo

- Led 5 independent laboratory sections per semester across 4 consecutive years, guiding 200+ students per semester through hands-on tensile and hardness testing, heat treatment and quenching studies, metallographic sample preparation, optical microscopy, and microstructural analysis.
- Ran pre-lab briefings and equipment demonstrations; supervised student operation of mechanical test frames and sample-preparation equipment; graded and gave written feedback on formal laboratory reports.
- Coached students on data reduction, uncertainty treatment, and technical writing, for many students the first formal engineering report of their degree.

Teaching Assistant — MAE 381, Engineering Materials *Fall 2022 to 2025*

Department of Mechanical and Aerospace Engineering, University at Buffalo

- Supported 300 students per semester through office hours, grading, and problem-session instruction on crystal structure, point and line defects, diffusion, binary phase diagrams, phase transformations, strengthening mechanisms, and mechanical behavior.

Teaching Development — University at Buffalo *Fall 2025*

- Participated in the TA Pedagogy Training program, studying Universal Design for Learning and approaches to addressing barriers in engineering education.

RESEARCH EXPERIENCE

Graduate Research Assistant — University at Buffalo *Jan. 2022 – May 2026*

Multiscale/Multiphysics Theory and Simulation (MulTS) Laboratory

- Explained the competitive-adsorption mechanism behind moisture-driven CO_2 capacity loss in zeolite 13X by simulating mixed CO_2/water transport across a range of humidity levels, identifying site blocking as the dominant molecular mechanism.
- Investigated how surface chemistry and confinement govern pressure-driven water and CO_2 transport in silica nanochannels using molecular dynamics. Connected density, hydrogen bonding, and velocity profiles to flow suppression and enhancement; found more than an order-of-magnitude difference in apparent water permeability between hydroxylated and graphene-coated walls in a 6 nm channel.
- Combined Monte Carlo simulations with a modified Peng–Robinson equation of state to describe CO_2/n-heptane and methane/n-heptane phase behavior under confinement. Optimized parameters at sampled temperatures, compared predictions with bulk reference data, and quantified sampling uncertainty through a 25-run analysis.
- Built coupled MD/Monte Carlo simulation workflows for binary mixtures across a range of pressures, temperatures, and pore sizes, enabling molecular-level phase prediction inaccessible to bulk models.
- Developed Python/Bash workflows for LAMMPS and Towhee simulations, HPC job submission, data extraction, and visualization, supporting reproducible parameter studies and consistent post-processing.

Research Assistant — Imam Khomeini International University *Feb. 2018 – Nov. 2021*

Advanced Simulation and Computing Laboratory (ASCL)

- Quantified stress–strain response, Young's modulus, fracture strain, and deformation mechanisms across 8 MXene structures under tensile loading; identified how surface termination and oxidation alter stiffness trends between carbide and nitride variants.
- Developed defect-concentration models for shear modulus and fracture strain in monolayer black phosphorene, linking atomic-scale defect populations to mechanical degradation and predictive structural reliability.
- Computed interfacial thermal conductance across 15 carbon-based heterostructures using non-equilibrium MD, showing that localized interfacial vibrational modes are a stronger physics-based descriptor than phonon DOS overlap.
- Established density and Young's modulus as composite descriptors for thermal-conductivity screening across 15 graphene polymorphs, enabling rapid structure–property evaluation without simulating every candidate.
- Trained moment tensor interatomic potentials using active learning on sparse DFT reference datasets for thermomechanical property calculations.

PUBLICATIONS

Refereed Journal Articles

1. **Hatamlee SM**, Hoepfner M, Elmehlawy M, Lu M, Li Z, Li J. Monte Carlo Informed Modification of Peng-Robinson Equation of State for CO_2/Hydrocarbon Mixtures Confined in Nanopores. *Accepted, Langmuir*, 2026.

2. **Hatamlee SM**, Li J. Molecular dynamics study of fluid flow in silica nanochannels to understand the transition from flow enhancement to reduction. *Journal of Molecular Liquids*, 2026:129795.

3. Xia Y, Liu J, Kancharla R, Li J, **Hatamlee SM**, Ren G, Semeykina V, Hamed A, Kane JJ. Insights into the 3D permeable pore structure within novel monodisperse mesoporous silica nanoparticles by cryogenic electron tomography. *Nanoscale Advances*, 5(11):2879–2886, 2023.

4. Ghasemi H, **Hatamlee SM**, Tirkolaei HK, Yazdani H. Biocementation of soils via enzyme-induced carbonate precipitation: an integrated laboratory and molecular dynamics study. *Biophysical Chemistry*, 284:106793, 2022.

5. **Hatamlee SM**, Jabbari F, Rajabpour A. Interfacial thermal conductance between gold and SiO_2: a molecular dynamics study. *Nanoscale and Microscale Thermophysical Engineering*, 2022.

6. **Hatamlee SM**, Gordiz K, Rajabpour A. Lattice-dynamics-based descriptors for interfacial heat transfer across 2D carbon-based nanostructures. *Journal of Applied Physics*, 2021.

7. Arabha S, Aghbolagh ZS, Ghorbani K, **Hatamlee SM**, Rajabpour A. Recent advances in lattice thermal conductivity calculation using machine-learning interatomic potentials. Journal of Applied Physics, 130(21):210903, 2021.

8. **Hatamlee SM**, Esfandiar A, Rajabpour A. Mechanical behaviors of titanium nitride and carbide MXenes: a molecular dynamics study. *Applied Surface Science*, 2021.

9. **Hatamlee SM**, Peer-Mohammadi H, Rajabpour A. Tuning shear mechanical properties and tensile strength anisotropy of monolayer black phosphorene: a molecular dynamics study. *Materials Today Communications*, 26 (2021), 101796.

10. **Hatamlee SM**, Rajabpour A, Volz S. Thermal conductivity of graphene polymorphs and compounds: from C_3N to graphdiyne lattices. *Carbon*, 161:816–826, 2020.

Manuscripts Under Review

11. **Hatamlee SM**, Li J. Molecular insights into water and CO_2 transport and adsorption in zeolite 13X. *Under review, Microporous and Mesoporous Materials*, 2026.

CONFERENCE PRESENTATIONS

Hatamlee SM, Li J. Molecular dynamics study of water flow in silica nanochannels: surface wettability governs the transition from reduction to enhancement. American Physical Society, Division of Fluid Dynamics, 78th Annual Meeting, Nov. 2025.

Hatamlee SM, Li J. Modified Peng–Robinson equation of state for CO_2/hydrocarbon binaries under confinement based on Monte Carlo simulations. MUSE All-Hands Meeting, University of Utah, Salt Lake City, Jan. 2024.

Hatamlee SM, Rajabpour A, Volz S. Thermal conductivity of graphene polymorphs and compounds: from C_3N to graphdiyne lattices. 7th National Conference on Nanotechnology from Theory to Application, Jun. 2019.

TECHNICAL SKILLS

Simulation methods: Molecular dynamics (equilibrium and non-equilibrium); Monte Carlo (GCMC, GEMC); density functional theory; machine-learning interatomic potentials; equation-of-state and continuum modeling.

Software: LAMMPS, Towhee, VASP, Quantum ESPRESSO, OVITO, VMD, COMSOL, Abaqus.

Programming: Python (NumPy, SciPy, Pandas, Matplotlib, scikit-learn), MATLAB, Bash, LaTeX.

Computing: High-performance computing (SLURM), Linux, Git, automated simulation workflows.

Laboratory instruction: Uniaxial tensile and hardness testing, heat treatment, metallographic preparation, optical microscopy, microstructural analysis.

AWARDS, HONORS, AND CERTIFICATES

2025 J. & S. Niciszewska Mucha Endowed Graduate Scholarship, University at Buffalo ($10,000)

2024 Leadership Grant, Council of Advocacy and Leadership Programming, University at Buffalo ($6,500)

2013 Ranked 738 among 250,000+ participants, Iranian University Entrance Exam

2019 Certificate: A Not-so-brief Introduction to Molecular Biophysics, IPM, Tehran

2018 Certificate: Quantum Calculation using Quantum ESPRESSO, Sharif University of Technology

2018 Certificate: 1st Conference on the Computation of Nanostructures, Sharif University of Technology

SERVICE AND LEADERSHIP

2024–2025 President, Mechanical and Aerospace Engineering Graduate Student Association, University at Buffalo; organized professional-development and community events; represented graduate students to department leadership.

2023–2024 Chair, Graduate Student Association International Committee, University at Buffalo.

2022–2024 President, Iranian Graduate Student Association, University at Buffalo — built orientation, peer-mentoring, and community-support programming for incoming international graduate students.

2022, 2024 Peer reviewer, Journal of Applied Physics (3 manuscripts).